Institution's Innovation Council
MHRD's Innovation Cell, AICTE

Innovation / Prototype
(Product/Service/Process)

Team Details

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Title (20 Words Max)	Speech Emotion Recognition Using Mfcc and Rnn
***Developed as part of**	***Academic Requirement/Study Project***
***Choose the Financial Year, during the Idea-PoC/Innovation Developed**	2025-26
***Sector / Domain**	Software - Web App Development
Innovation Type	Process, Service

***Development Stage - Technology Maturity of the Solution/Innovation in terms of Technology Readiness Level TRL (if applicable (Refer TRL Stages)**	TRL 7 : Demonstration system operating in operational environment at pre-commercial scale
Development Stage - Manufacturing Maturity of the Solution/Innovation in terms of Manufacturing Readiness Level (MRL) Refer MRL Stages	MRL 3: Manufacturing proof of concept developed
Development Stage: Investment Readiness Level of the Solution/Innovation (IRL). Refer IRL Stages	IRL 4: Prototype Low-Fidelity Minimum Viable Product (MVP): "Low-fidelity" - A representative of the component or system that has limited ability to provide anything but initial information about the end product.

***Define the problem and its relevance to today's market / society / industry need (Max: 100 Words)**	In today's digital era, most communication systems lack the ability to understand human emotions, leading to ineffective interactions in applications like customer support, virtual assistants, and mental health monitoring. This gap reduces user satisfaction and limits the effectiveness of AI-driven services. With the increasing reliance on voice-based technologies, there is a growing need for systems that can interpret emotional context from speech. Addressing this need, the proposed solution focuses on recognizing emotions from voice data, enabling more intelligent, responsive, and human-centric systems across industries such as healthcare, customer service, and smart technologies.
***Describe the Solution / Proposed / Developed (Max: 100 Words)**	The proposed solution is a Speech Emotion Recognition system that identifies human emotions from voice inputs. It uses MFCC for extracting audio features and a Recurrent Neural Network (RNN) for accurate classification of emotions. The system is implemented as a web application using Flask, allowing users to upload or record audio and receive real-time emotion predictions with confidence scores. Trained on labeled datasets, the model ensures reliable performance. This solution enhances human-computer interaction by enabling emotion-aware systems, making it useful for applications in healthcare, customer service, and intelligent virtual assistants.
***Explain the uniqueness and distinctive features of the (product / process / service) solution (Max: 100 Words)**	The proposed solution is unique in its integration of MFCC-based feature extraction with a deep learning RNN model to achieve accurate and efficient emotion detection from speech. Unlike many existing systems, it offers real-time predictions through a lightweight and user-friendly web interface, requiring no specialized hardware. The system is scalable, adaptable to different voice inputs, and designed for practical deployment. Its ability to combine accuracy, accessibility, and real-time processing makes it distinctive, enabling emotion-aware interactions across applications such as healthcare, customer service, and intelligent virtual assistants.
***How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any (Max: 100 Words)**	Compared to existing solutions, this system uses a deep learning RNN model combined with MFCC features to achieve higher accuracy in emotion detection from speech. Many competitors rely on traditional machine learning methods or complex setups, whereas this solution is designed as a lightweight, web-based application that requires no specialized hardware. It provides real-time predictions with confidence scores, making it more practical and user-friendly. Additionally, it focuses on easy integration and scalability, allowing deployment across multiple domains such as healthcare, customer service, and AI-based applications.
Is there any IP or Patentable Component associated with the Solution?	NO

Has the Solution Received any Innovation Grant/Seed fund Support?	NO
Are there any Recognitions (National/International) Obtained by the Solution?	NO
Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?	NO
Had the Solution Received any Pre-Incubation/Incubation Support?	NO
***Utility: Highlight the utility/value proposition (key benefits) aspects of the solution/innovation* (Max: 100 Words)**	The solution delivers high value by enabling real-time emotion detection from speech, improving the quality of human-computer interaction. It helps organizations understand user sentiment, enhancing customer support and decision-making. In healthcare, it can assist in monitoring emotional well-being, while in education and virtual assistants, it enables more personalized and responsive systems. The solution is cost-effective, easy to deploy, and accessible through a web interface, making it suitable for widespread adoption. Its ability to provide accurate, scalable, and practical emotion insights makes it highly beneficial across multiple industries.
***Scalability: Highlight the market potential aspects of the Solution/Innovation (Potential Market Size, segmentation and Target users/customers etc.) (Max: 100 Words)**	The solution has strong scalability with applications across healthcare, customer service, education, and AI-driven industries. The global rise in voice-based technologies and emotion-aware AI creates a large and growing market opportunity. Target users include call centers, healthcare providers, edtech platforms, and enterprises integrating AI solutions. Its web-based architecture enables easy deployment and expansion across regions with minimal infrastructure cost. The solution can be offered as a scalable service or integrated via APIs, allowing it to reach diverse market segments and support large user bases, making it commercially and technically scalable.
***Economic Sustainability: Highlight commercialisation/business application aspects of the solution (how it is going to economic profitable and viable) (Max: 100 Words)**	The solution is economically sustainable as it can be commercialized as a SaaS platform offering emotion recognition services to enterprises. Businesses such as call centers, healthcare providers, and AI solution companies can subscribe for real-time emotion analysis. Revenue can be generated through subscription plans, API-based usage, and licensing. Its low deployment and maintenance costs ensure higher profit margins. With the increasing demand for AI-driven insights, the solution offers strong opportunities for scalability, recurring revenue, and long-term profitability across multiple industries.

***Environmental Sustainability: Highlight environmental friendliness aspects and related benefit of the solution/innovation (Max: 100 Words)**	The solution promotes environmental sustainability as a fully digital, software-based system that reduces reliance on physical resources and paper-based processes. Its web-based deployment supports remote usage, minimizing travel and associated carbon emissions. The lightweight design ensures efficient use of computational resources, reducing energy consumption compared to large-scale systems. By enabling automated analysis and digital interaction, it contributes to eco-friendly practices and supports sustainable technological adoption with minimal environmental impact.
Video URL	https://drive.google.com/file/d/1Vty2f0MchVJ5OLaGV-Akm--pHmFUIHW2/view?usp=sharing
Upload Photograph: (JPG, PNG, PDF max 2 MB)	https://drive.google.com/file/d/1b7SRI57nRO0-ZfQvSQmxi7om3EFuTQX1/view?usp=sharing

(e) Timeline- Develop a timeline against the milestones for taking Idea/PoC to Prototype Development and (or) Commercial level/start-up stage.	Idea & Research – 1 month Data Collection & Model Design – 2 months Model Development & Testing – 2 months Web App Development – 1 month Prototype Completion – 1 month Future: Commercialization – 6–12 months
Applicability of Solution 10 Marks *(Max 50 words for each from a-e)*	
(a) **Usability:** what is the usability of your innovation. *(Level of acceptance of innovation and its Features among target group)*	The solution offers high usability with a simple web interface for uploading or recording audio and getting real-time emotion predictions. It requires no technical expertise, provides clear outputs, and is easily accessible. Its practical applications across healthcare, customer service, and education ensure strong acceptance among target users.
(b) **Scalability:** how your innovation will be scalable at market level.	The innovation is highly scalable due to its web-based architecture, enabling deployment across multiple regions with minimal infrastructure. It can serve diverse industries such as healthcare, customer service, and education. The solution can be expanded via APIs and cloud integration, allowing it to handle large user bases and adapt to growing market demands.
(c) **Economic sustainability:** Explain the potential of innovation to become profitable or financially viable.	The innovation has strong economic potential as it can be offered as a subscription-based or API-driven service to businesses. Its low deployment and maintenance costs ensure profitability. With increasing demand for AI-driven emotion analysis in sectors like healthcare and customer service, it supports recurring revenue and long-term financial sustainability.
(d) **Environment Sustainability:** How your innovation is environment friendly or address environmental problems.	The innovation is environmentally friendly as it is a fully digital solution that reduces the need for physical resources and paper usage. Its web-based deployment supports remote access, minimizing travel and carbon emissions. The efficient design ensures low energy consumption, promoting sustainable and eco-friendly technology adoption.
(e) Is there any Intellectual Property (IP) Component associated with innovation? if yes, explain.	Currently, there is no registered Intellectual Property associated with the innovation, as it uses established techniques like MFCC and RNN. However, the system's unique integration, real-time processing, and deployment approach offer potential for future IP protection or patent filing after further development and optimization.
Define the potential market size (in terms of INR) and target customers. *(Max 100 words)*	The potential market size for emotion recognition and AI-based voice analytics in India is estimated to exceed ₹5,000–₹8,000 crore, driven by growth in customer service, healthcare, and AI-driven applications. Target customers include call centers, healthcare providers, edtech platforms, enterprises, and AI solution developers. With increasing adoption of voice-based technologies, the solution can serve both B2B clients and scalable digital platforms, offering significant market

	opportunities and expansion potential.

INSTITUTION'S INNOVATION COUNCIL (Ministry of HRD Initiative)

Annexure 1

Themes:

1. Healthcare & Biomedical devices.
2. Agriculture & Rural Development.
3. Smart Vehicles/ Electric vehicle/ Electric vehicle motor and battery technology.
4. Food Processing.
5. Robotics and Drones.
6. Waste management.
7. Clean & Potable water.
8. Renewable and affordable Energy.
9. IoT based technologies (e.g. Security & Surveillance systems etc)
10. ICT, cyber physical systems, Block chain, Cognitive computing, Cloud computing, AI & ML.

10 stages of MRL

MRL 1: Basic manufacturing implications identified
MRL 2: Manufacturing concepts identified
MRL 3: Manufacturing proof of concept developed
MRL 4: Capability to produce the technology in a laboratory environment
MRL 5: Capability to produce prototype components in a production relevant environment
MRL 6: Capability to produce a prototype system or subsystem in a production relevant environment
MRL 7: Capability to produce systems, subsystems or components in a production representative environment.
MRL 8: Pilot line capability demonstrated. Ready to begin low rate production.
MRL 9: Low rate production demonstrated. Capability in place to begin Full Rate Production.
MRL 10: Full rate production demonstrated and lean production practices in place.

9 stages of TRL:

TRL 0 : Idea. Unproven concept, no testing has been performed.
TRL 1 : Basic research. Principles postulated observed but no experimental proof available.
TRL 2 : Technology formulation. Concept and application have been formulated.
TRL 3 : Applied research. First laboratory tests completed; proof of concept.
TRL 4 : Small scale prototype built in a laboratory environment ("ugly" prototype). TRL 5 : Large scale prototype tested in intended environment.
TRL 6 : Prototype system tested in intended environment close to expected performance. TRL 7 : Demonstration system operating in operational environment at pre-commercial scale. TRL 8 : First of a kind commercial system. Manufacturing issues solved.
TRL 9 : Full commercial application, technology available for consumers.

INSTITUTE READINESS LEVEL

IRL 1: Basic Research (Need Identification & Peer Review Publications) & Completed First-Pass Business Model Canvas (BMC)

IRL 2: Applied Research (Market Size and Competitive Analysis) & Business Plan – Value Proposition & IP Identification

IRL 3: Validate Problem - Solution Fit (Confirmed Value Proposition & Techno-Economic Analysis) & Minimum Product Cost (Maturity of Core Technology)

IRL 4: Prototype Low-Fidelity Minimum Viable Product (MVP): "Low-fidelity" - A representative of the component or system that has limited ability to provide anything but initial information about the end product.

IRL 5: Validate Product-Market Fit (Integrated Validation of the Minimum Viable Process and Process Engineering). "High-fidelity" - A high-fidelity laboratory environment would involve testing with equipment that can simulate and validate all system specifications within a laboratory setting.

IRL 6: Validate Business/Revenue Model: Integrated Pilot Development– understanding operational nuances

IRL 7: Prototype High Fidelity MVP: Integrated Pilot Continuous Operation

IRL 8: Pre-Commercial Demonstration – Operating Conditions and quality stabilized

IRL 9: Full Commercial Development – A full time process engineering staff continuously verifies that operations are meeting cost, yield and productivity targets.